

The total possible points are 47:

P1 (20): DOF worth 5 points; each of equations worth 1 point and in total 5 points; right solution for the values in the equation system worth 5 points; molar flow rates, compositions for fresh feed, compositions for recycle stream each worth 2 points and in total 6 points.

P2 (12): flow rate for the final product worth 4 points; the composition of the recycled stream worth 4 points; fractions of ethane and methane worth 4 points.

P3 (14): right answer for the temperature worth 4 points; solutions for liquid and vapor each worth 2 points; fractional recovery of ethanol, fractional recovery of water and separation factor each worth 2 points.